

NARWASTU DWI NILAN BARA' ALLO

Denpasar, Bali, Indonesia | [+62 813-3244-4081](tel:+6281332444081) | dwinarwastu02@gmail.com
[linkedin.com/in/dwinarwastu](https://www.linkedin.com/in/dwinarwastu) | github.com/dwinarwastu | dwinn-studio.my.id

WORK EXPERIENCE

Programmer (Backend Developer)

Aug 2025 – Present

PT. Semua Aplikasi Indonesia

Denpasar, Bali

- Architected and engineered the core multi-tenant backend for **NontonPlus V2** (Hospitality IPTV platform) from scratch using **NestJS**, **PostgreSQL**, and **MongoDB**.
- Implemented low-latency bidirectional real-time telemetry and device communication protocols using **Socket.IO**.
- Engineered a scalable multi-tenant management service tailored for ISP clients with strict tenant-level data isolation.
- Reduced database read-overhead and optimized session handling by implementing high-performance caching layers via **Redis**.
- Integrated object storage with **MinIO** and established end-to-end system health observability and log aggregation using **Grafana** and **Loki**.

Programmer Intern

Feb 2025 – Jun 2025

PT. Semua Aplikasi Indonesia

Denpasar, Bali

- Developed backend RESTful APIs for inpatient survey modules and real-time emergency nurse-call features on the **Nonton+ Hospital** application using **Node.js** and **Socket.IO**.
- Collaborated with cross-functional teams to test socket connection reliability and telemetry payload parsing across hospital room devices.

Programmer Intern

Jun 2024 – Aug 2024

CV. Natusi

Mojokerto, Jawa Timur

- Developed the digital medication and fluid administration tracking module for the emergency room (IGD) website of **RSUD Dr. Wahidin Sudiro Husodo**.
- Streamlined clinical data entry workflows, reducing manual documentation time for medical personnel using **Laravel** and **MySQL**.

Programmer Intern

Jul 2020 – Sep 2020

PT. Anekapay Teknologi Indonesia

Kediri, Jawa Timur

- Developed a concurrent web scraper using **Go (Golang)** to extract product catalogs, pricing, and merchant data from e-commerce platforms.
- Structured and parsed raw JSON marketplace data pipelines for downstream payment and commerce workflows.

KEY PROJECTS

Interactive API Playground & FinTech Gateway

2026

Go, Chi Router, SQLite WAL, React, TypeScript

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- Engineered an enterprise developer portfolio and single-binary server showcasing Go sub-millisecond telemetry and Clean Architecture.
- Implemented a zero-allocation Webhook HMAC-SHA256 signature verification engine and a Token Bucket rate-limiting simulator with sub-ms atomic overhead.

Hospital Nurse-Call & Emergency Telemetry

2025

Node.js, Express, Socket.IO, MongoDB

Denpasar, Bali

- Engineered a real-time event-driven emergency signaling backend between inpatient hospital rooms and on-duty nursing stations.
- Implemented low-latency socket room clustering and automated failover message delivery protocols for critical patient care.

EDUCATION

Politeknik Negeri Malang

2022 – 2025

Associate Degree (D3) in Informatics Management

Malang, Jawa Timur

SMK Telkom Malang

2019 – 2022

Vocational High School Diploma in Software Engineering

Malang, Jawa Timur

TECHNICAL SKILLS

Languages & Backend: Go (Golang), TypeScript, JavaScript, Node.js, NestJS, Express, PHP, Laravel, RESTful APIs

Databases & Caching: PostgreSQL, MongoDB, SQLite (WAL Mode), Redis

Architecture & Real-Time: Multi-Tenant Systems, WebSockets, Socket.IO, Webhook HMAC Verification, Clean Architecture

DevOps & Observability: Docker, Git, MinIO (S3-compatible), Grafana, Loki, Postman, Linux

Languages: Indonesian (Native), English (Technical / Working Proficiency)